Methods in genome annotation

with a focus on eukaryotes and protein coding genes

22/09/2025

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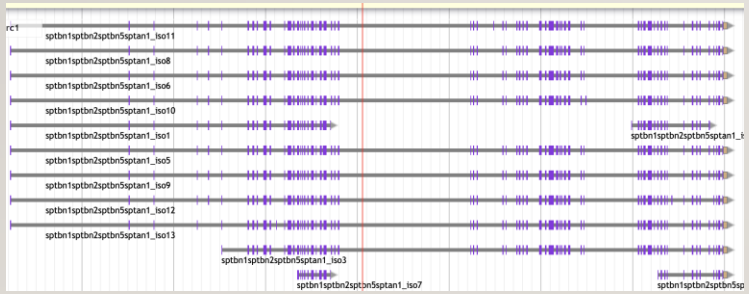


What is annotation ?

Structural annotation:

Find out where the regions of interest (usually genes) are in the sequence data and what they look like.

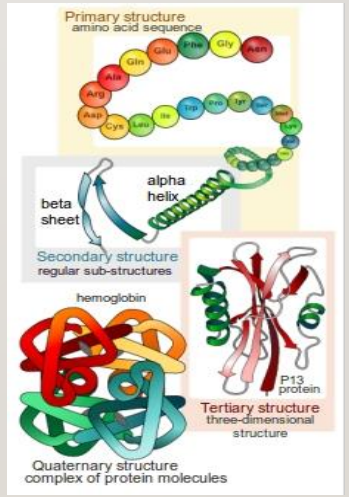
=> **Gene prediction** / **Gene Finding**



vs

functional annotation:

Find out what the regions do.
What do they code for?





Annotation tools

First annotation tools

90's

>100 annotation tools*

Nowadays

Structural annotation + functional annotation + …

GFF

Experimental vs Preditive

Experimental
(ESTs, cDNAs, RNA-seq)

Isolate and clone cognate transcripts (as cDNA) sequence them and compare cDNA with genomic DNA

=> It's the ONLY secure method but:

- Cloning is time consuming.
- Lowly expressed genes are difficult to detect.
- …

Predictive

- Intrinsic (Machine Learning)

Use only the DNA sequence itself to predict genes based on statistical models of gene structure

- Extrinsic

Use **external biological evidence**, like RNA-Seq, ESTs, protein alignments, or known homologs, to guide annotation

- Hybrid

Modern methods often integrate intrinsic and extrinsic signals, blurring the line…

The different approaches

- Similarity-based methods :

  These use similarity to annotated sequences like proteins and transcripts

- *Ab initio* prediction :

  Machine learning based methods (probabilistic: Likelihood, Bayesian | deterministic: Neural networks)

- Hybrid approaches :

  Ab initio tools with the ability to integrate external evidence/hints

- Comparative (homology) based gene finders :

  These align genomic sequences from different species and use the alignments to guide the gene predictions

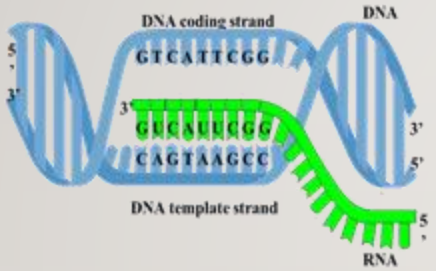
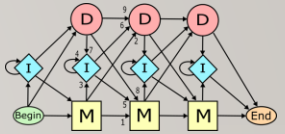
- Chooser, combiner approaches :

  These combine gene predictions of other gene finders

- Pipelines :

  These combine multiple approaches, can annotate other types of features, work on functional annotation, etc.

Types data used vs methods - the main lines

Annotation approach	Extrinsic			Intrinsic
	Transcripts	Proteins	Genomes	ML models
	Assembled from RNA-seq or downloaded ESTs 	Known amino acid sequences from other organisms 	Genome from close related species 	probabilistic (HMM, GHMM)  deterministic (CNN, RNN i.e deep learning) 
Similarity	X	X		
Comparative			X	
Pure ab initio				X
Hybrid	X	X	X	X
Chooser/combiner	X	X		X
Pipeline	X	X		X

Extrinsic approaches

Similarity-based methods

These use similarity to annotated sequences like

Proteins

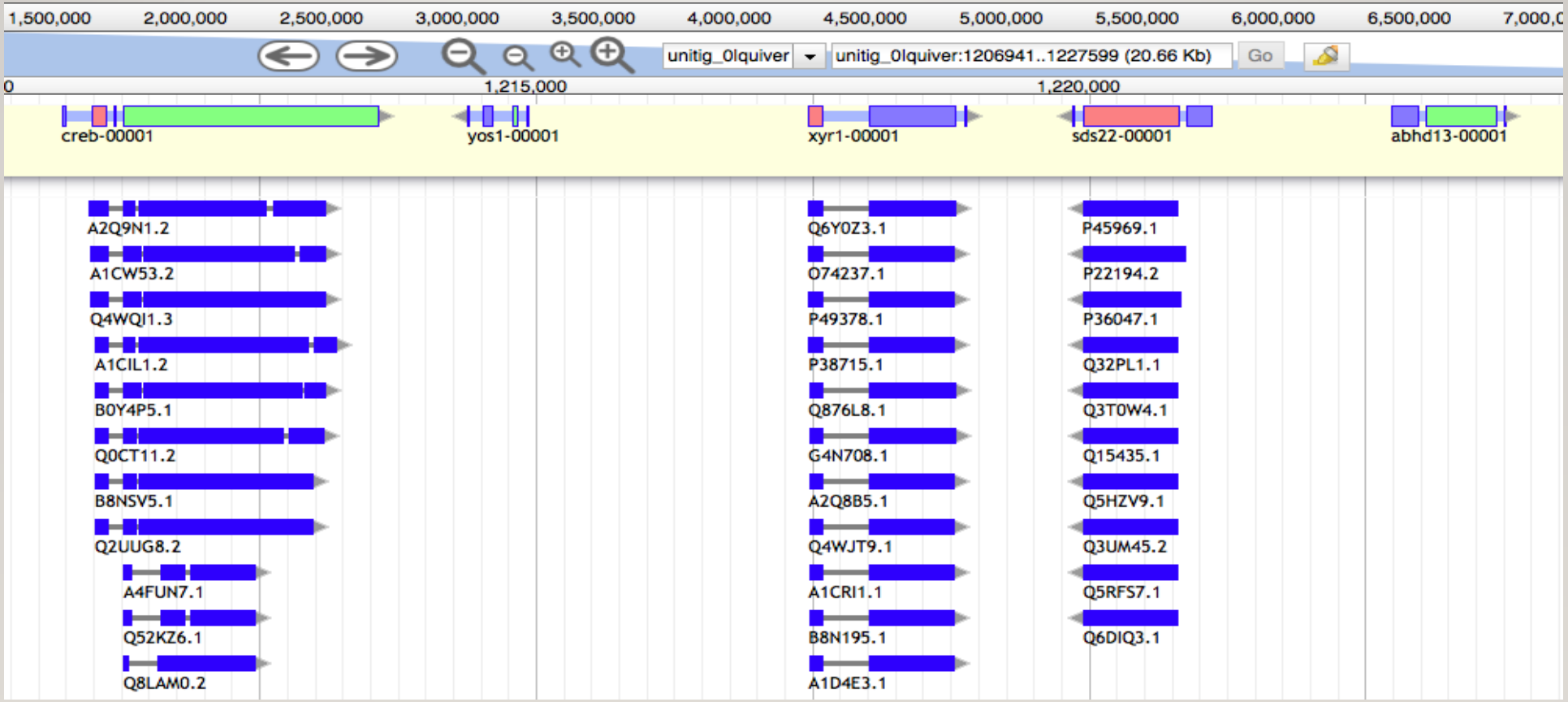
Transcripts

ESTs

Similarity-based method: Protein data

- Protein sequences are aligned to the genome

Conserved in sequence => conserved annotation with little noise



Similarity-based method: Protein data

Limits :

- Related to pre-existing data
- Proteins from model organisms often used => bias?
- Proteins can be incomplete
- Protein can be wrong (PE)
- No UTR

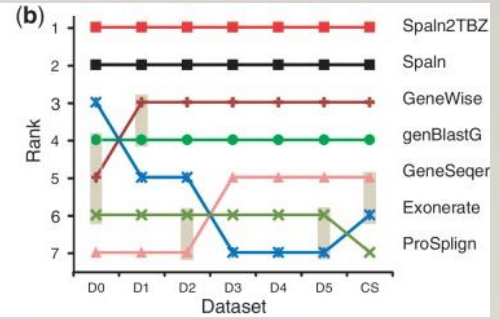
- Rough approximation (fast?)

AA
Pmatch
tblastn
PSimScan

- Splice-site aware alignment (slow?)

AA
Exonerate
Genewise
GenomeThreader
Miniprot

	Method	Human	
		Mouse	Chicken
protein	Exonerate	12h 36min 10.3s	7h 33min 17.0s
	genBlastG	3min 30.3s	2min 16.6s
	GeneSeqer	10h 10min 24.0s	6h 20min 40.0s
	GeneWise	69h 17min 36.1s	47h 36min 6.6s
	ProSplign	2h 18min 24.9s	1h 17min 39.1s
	Spaln2TBZ	4min 32.0s	4min 41.8s



Nucleic Acids Res. 2012 Nov 1;40(20):e161.
doi: 10.1093/nar/gks708. Epub 2012 Jul 30.

Popular tools

GenomeThreader

Exonerate

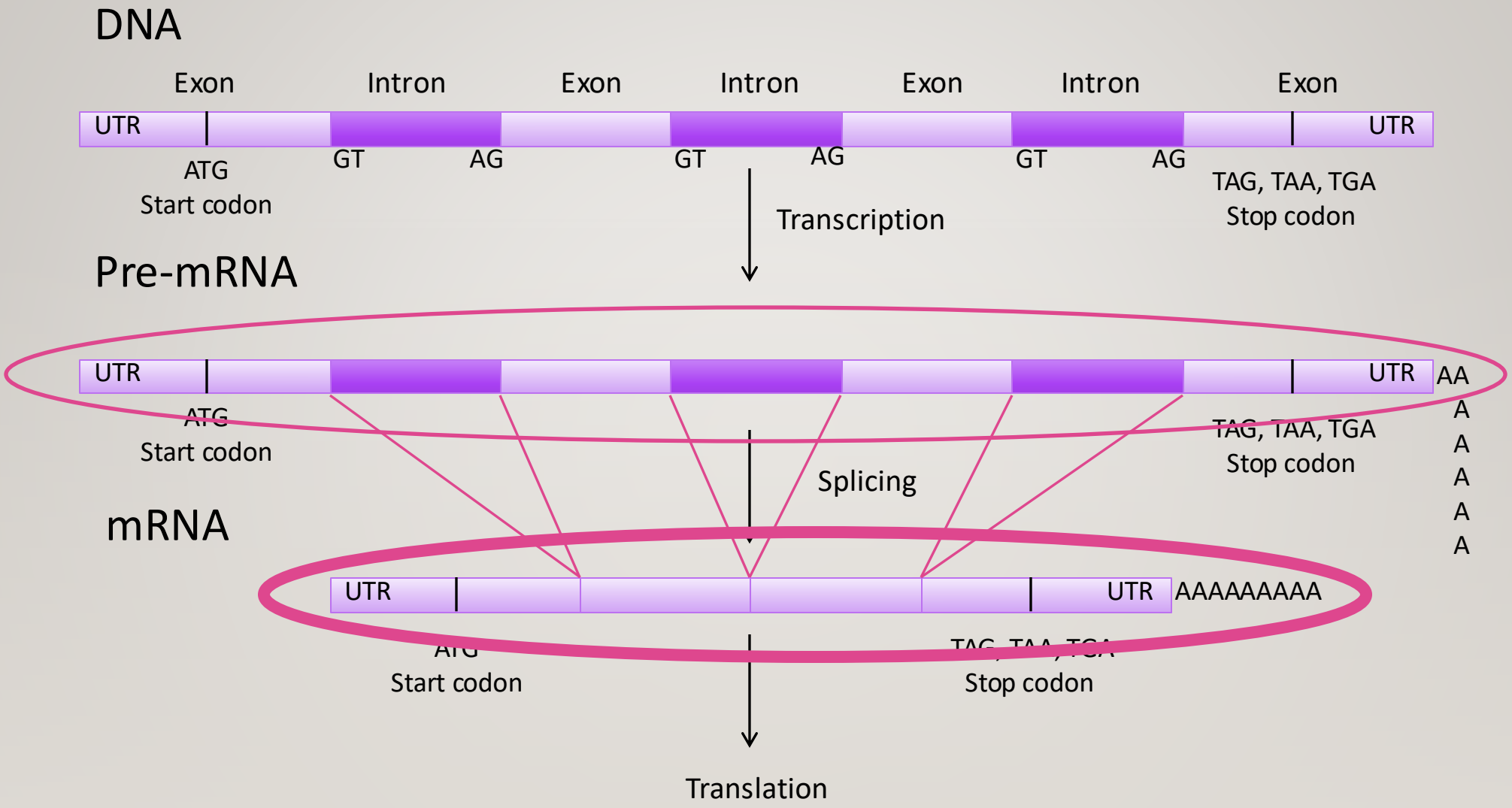
MAKER3



<=> BLASTX
GenomeThreader or Exonerate

Exercise

Similarity-based method: Transcript data





Short Reads

Long Reads

Guided Assembly



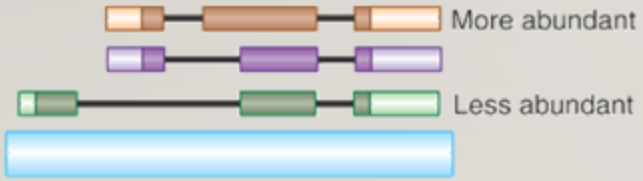
Hisat2, Star, Bowtie2, bwa-mem

Align reads to genome



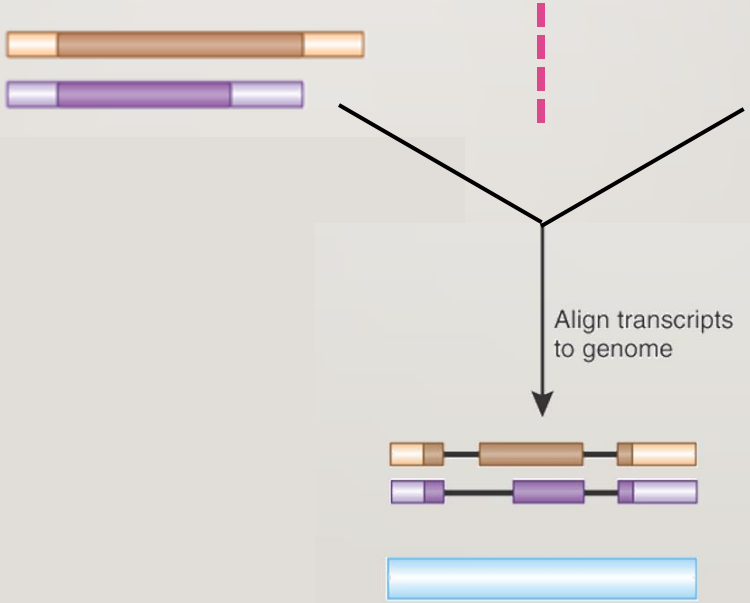
Assemble transcripts from spliced alignments

Stringtie



De novo Assembly

Assemble transcripts *de novo*



Splice-site aware alignment (slow?)

DNA
Exonerate
Gmap
GenomeThreader
Minimap2

Rough approximation (fast?)

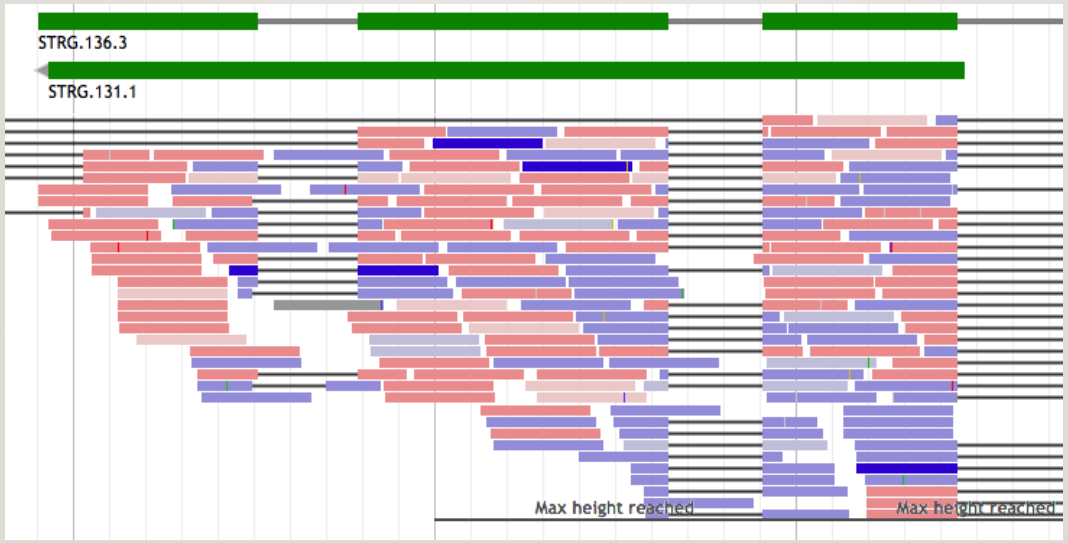
DNA
Blastn
Vsearch
NSimScan

https://home.cc.umanitoba.ca/~frist/PLNT7690/lec12/lec12.3.html

RNA-seq - Spliced reads



RNA-seq – pre-mRNA noise



Similarity-based method: Transcript data

- Should always be included in an annotation project
- From the same organism as the genomic data => unbiased
- Sample different tissues or life stages if possible

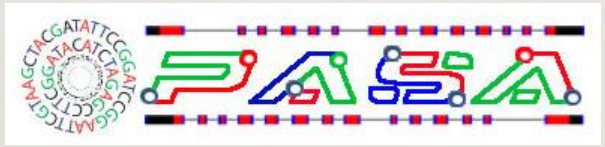
Limits :

- Hard to catch low expressed / peculiar expressed (stage of life, condition, etc.) / isoforms
- Can be very noisy (tissue/species dependent), can include pre-mRNA
- short-reads:
  - need to be assembled first
  - Transcriptome assembly errors
- Long-reads:
  - error rate / frameshift / indels

Popular tools

GenomeThreader

PASA



MAKER3



GMAP and/or BLAT <=>

BLASTN
GenomeThreader or Exonerate

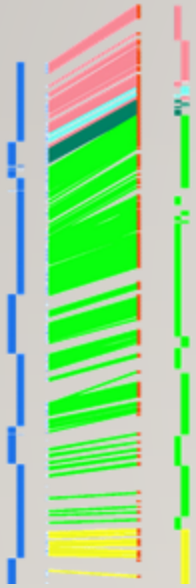
Exercise

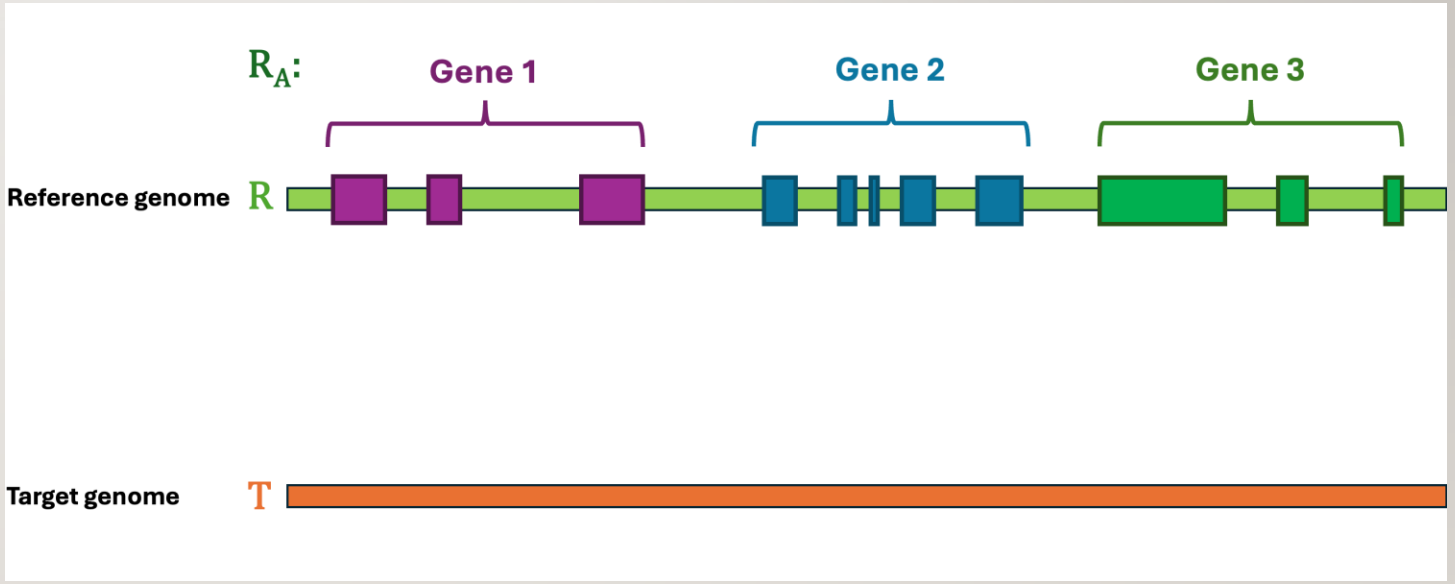
Extrinsic approaches

Comparative methods

Comparative-based method (projection tools)

Tools that use **whole-genome alignments** between closely related species.
Transfer gene models based on conserved synteny, gene structure, and alignment blocks.





LASTZ
Mummer
Satsuma
SynMap
} chained alignments

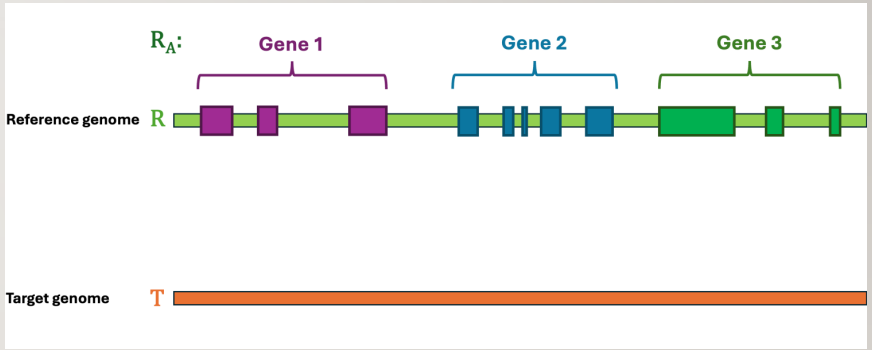
Pecan
HAL
} graph alignments

Popular tools

- **Kraken**
- **TransMap (UCSC)**

Comparative-based method (projection tools)

Tools that use **local alignments** between species.
Transfer gene models based on gene alignment.



Popular tools

LiftOff

transfer annotation between closely related species

minimap2

LiftOn

transfer annotation between close and distant related species

Miniprot

Exercise

Intrinsic approaches

ab initio

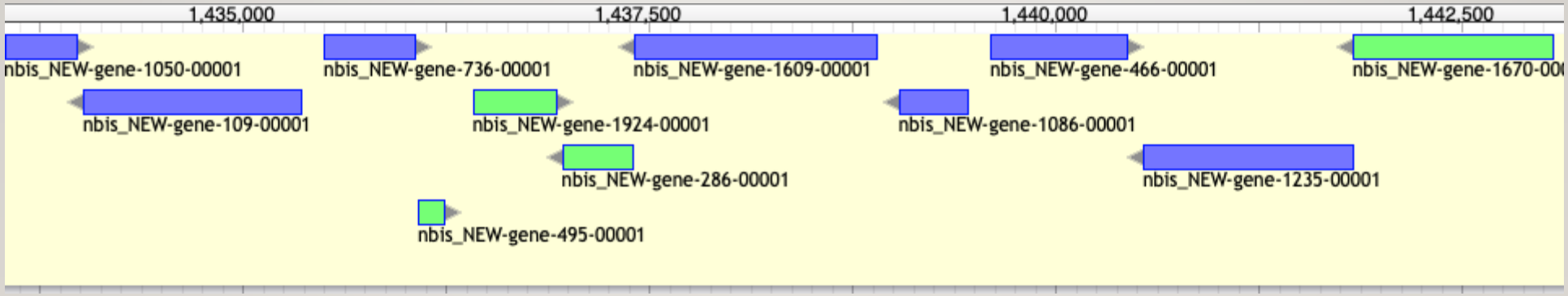
Using Machine Learning approaches (probabilistic and/or deterministic models) to predict features

- **Bacterial gene prediction**

  - Short intergenic regions

  - Uninterrupted ORFs (No intron)

  - Very conserved signals



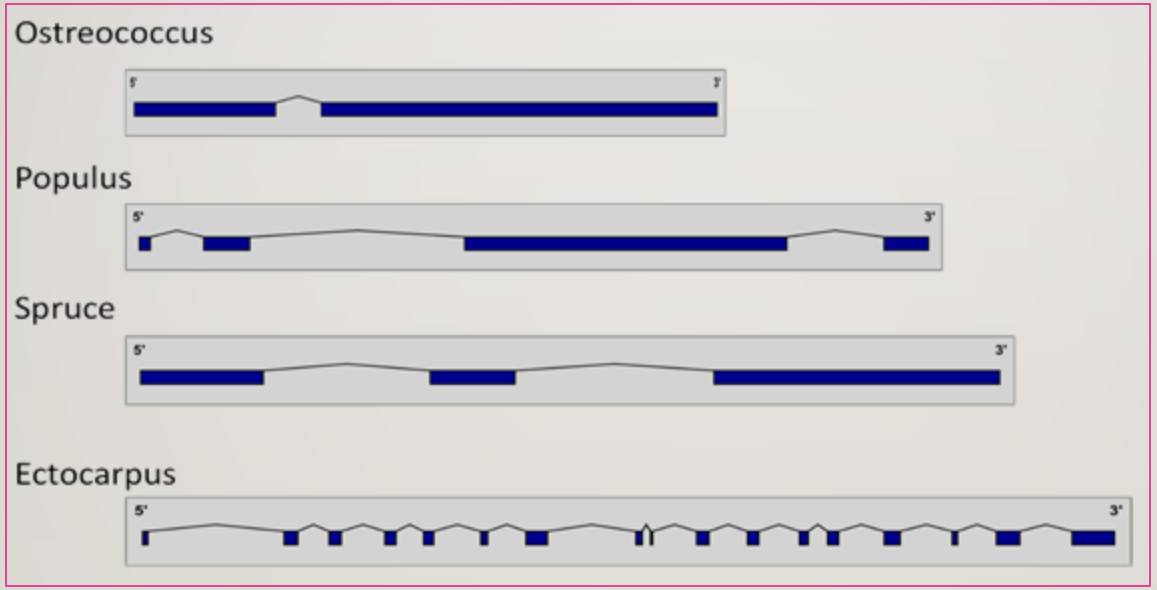
⇒ easy problem: Accuracy > 90%

- **Eukaryotic gene prediction**

  - Presence of intron => structures differ

  - Isoforms

  - New features (e.g. lncRNAs)



=> Hard problem: Accuracy < 70%

=> Requires training

Probabilistic methods

- **gene content**

Statistical properties of protein-coding sequence learn from existing data.

  - codon usage
  - hexamer usage
  - GC content
  - compositional bias between codon positions
  - nucleotide periodicity
  - exon/intron size
  - …

- **signal detection**

These are like **landmarks** in the genome, they tell the model *where* to look.

  - Promoter
  - ORF
  - Start codon
  - Splice site (Donor and acceptor)
  - Stop codon
  - Poly(A) tail
  - CpG islands
  - …

=> *Ab initio* tools will combine this information through different Probabilistic models (e.g. HMM, GHMM, WAM) to calculate the most probable combination that represents a valid gene

Popular tools

Self training!

- **SNAP** — Works ok, easy to train, not as good as others especially on longer intron genomes.
- **Augustus** — Works great, hard to train (but getting better).
- **GeneMark-ES** — no hints, buggy, not good for fragmented long introns (Best suited for Fungi).
- **FGENESH** — Works great, costs money even for training.

Supported by MAKER genomes or

http://weatherby.genetics.utah.edu/MAKER/wiki/index.php/MAKER_Tutorial

- **GlimmerHMM** (Eukaryote)
- **GenScan**
- **Gnomon** (NCBI)

Deterministic methods

Neural networks (CNN, bLSTM) predicts **for each base**, the probability of being:

- in an exon
- in an intron
- at a splice site
- in UTRs, etc.

+

HMM postprocessor

Captures complex, long-range sequence dependencies.
=> struggles to precisely locate exon boundaries
=> base-level predictions can be

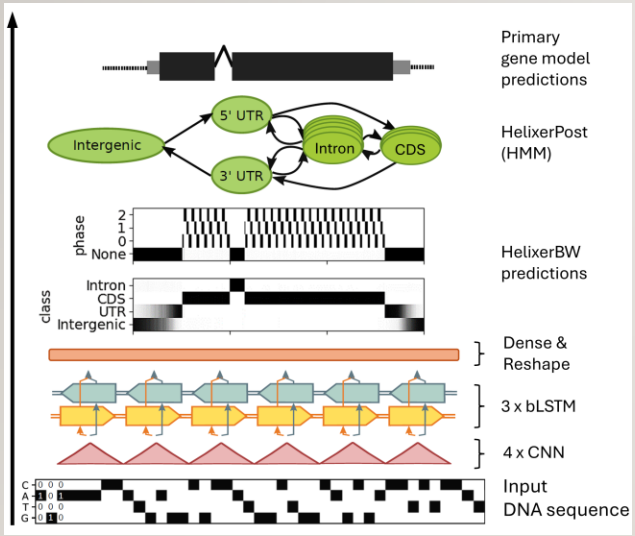
- **Noisy** (e.g., false positives),
- **Biologically inconsistent** (e.g., a stop codon in the middle of a predicted CDS),
- **Missing structure** (e.g., incorrect start/stop, fragmented exons).

Ensures coherent gene structures by filtering noisy outputs.

Popular tools

Helixer

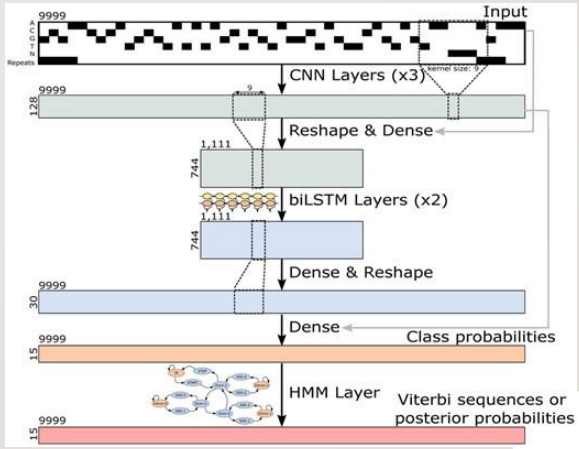
Helixer–de novo Prediction of Primary Eukaryotic Gene Models Combining Deep Learning and a Hidden Markov Model

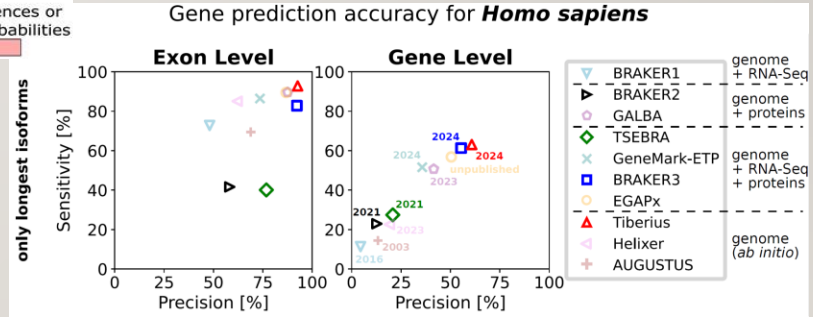


https://doi.org/10.1101/2023.02.06.527280
https://github.com/weberlab-hhu/Helixer

Tiberius

End-to-End Deep Learning with an HMM for Gene Prediction





https://academic.oup.com/bioinformatics/article/40/12/btae685/7903281
https://github.com/Gaius-Augustus/Tiberius

Training intrinsic / *ab initio* gene-finders

- *Ab-initio* tools need of a models (also called profile) => Training
  - Few self-trained tools, most need a separate training procedure
  - The quality of the gene-finder results hugely relies on the quality of the training!
  - Needs training for every genome / phylogenetic range (= different training sets)

A fungal genome

Fungi
Plants



Training

Probabilistic

A sets of high quality genes (>500)
=> These "known" genes are usually inferred from aligned transcripts or proteins

Deterministic

A sets of high quality genomes (>3)
=> Species that generalize well to your target species

Strengths :

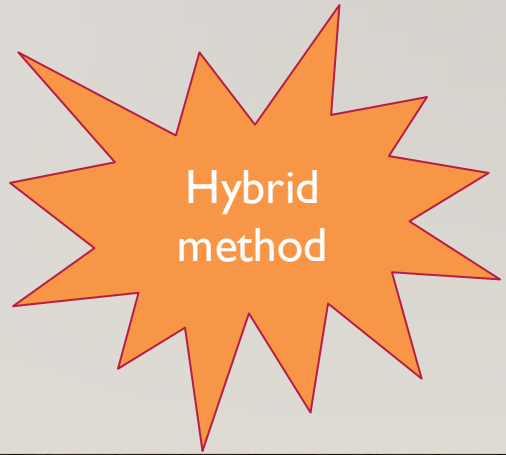
- Fast and easy
- Annotate unknown genes
- Sensitivity ok
- Need no external evidence**

Limits :

- No UTR*
- No alternatively spliced transcripts*
- Bad specificity / precision (Over prediction of exons or/and genes)
- **Training** needed (Need external evidence**)

Common errors in annotation:

- Split single gene into multiple predictions
- Fused with neighboring genes
- May be less accurate than homology based method:
  - Exon boundaries
  - Splicing sites



Exercise

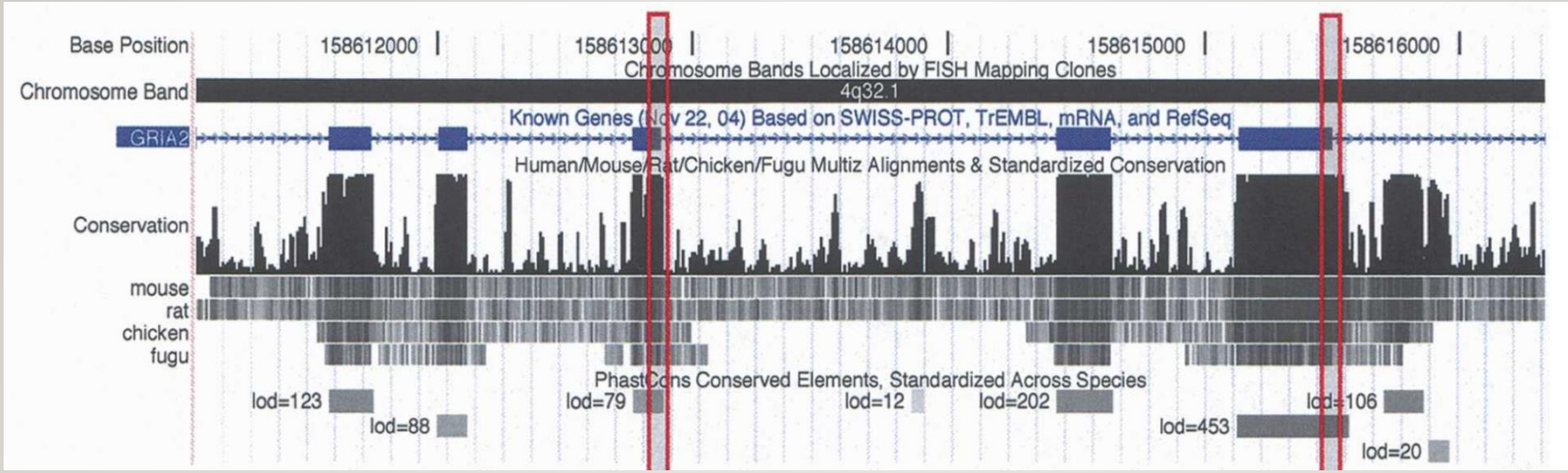
Hybrid approaches

Evidence-drivable gene predictors incorporate hints/extrinsic information in the form of alignments (transcript, protein, whole genome) to increase the accuracy of the gene prediction.

- *Can predict locus without extrinsic information*
- *Improve prediction when extrinsic information available*

Hybrid: Comparative + *ab initio*

functional parts e.g. exons, tend to be more conserved than the non-functional ones, the introns.



Alignment of genomic sequences from different species and use the alignments to guide the ab-initio gene predictions.

Limits : - Whole genome alignment is time/memory consuming
- Need relatively close related genome (<50 My)

Hybrid: Comparative + *ab initio*

- **Dual genome**, de novo gene structure prediction:
  - **Rosetta** (Pioneer – 2000)
  - **SGP-2** (2001) – considers only the conservation in protein-coding regions
  - **TWINSCAN** (2001) - included models of conservation in splice sites and start and stop codons
  - **SLAM** (2003)
  - **TWAIN** (2005)

- More than 2 genomic sequences:
  - **NSCAN*** (2006)
  - **Conrad** *(CRF, 2007)
  - **CONTRAST*** (CRF, 2008) -> 58% accuracy
  - **Augustus-CPG***

*Can use similarity-based information

Hybrid: Similarity based + *ab initio*

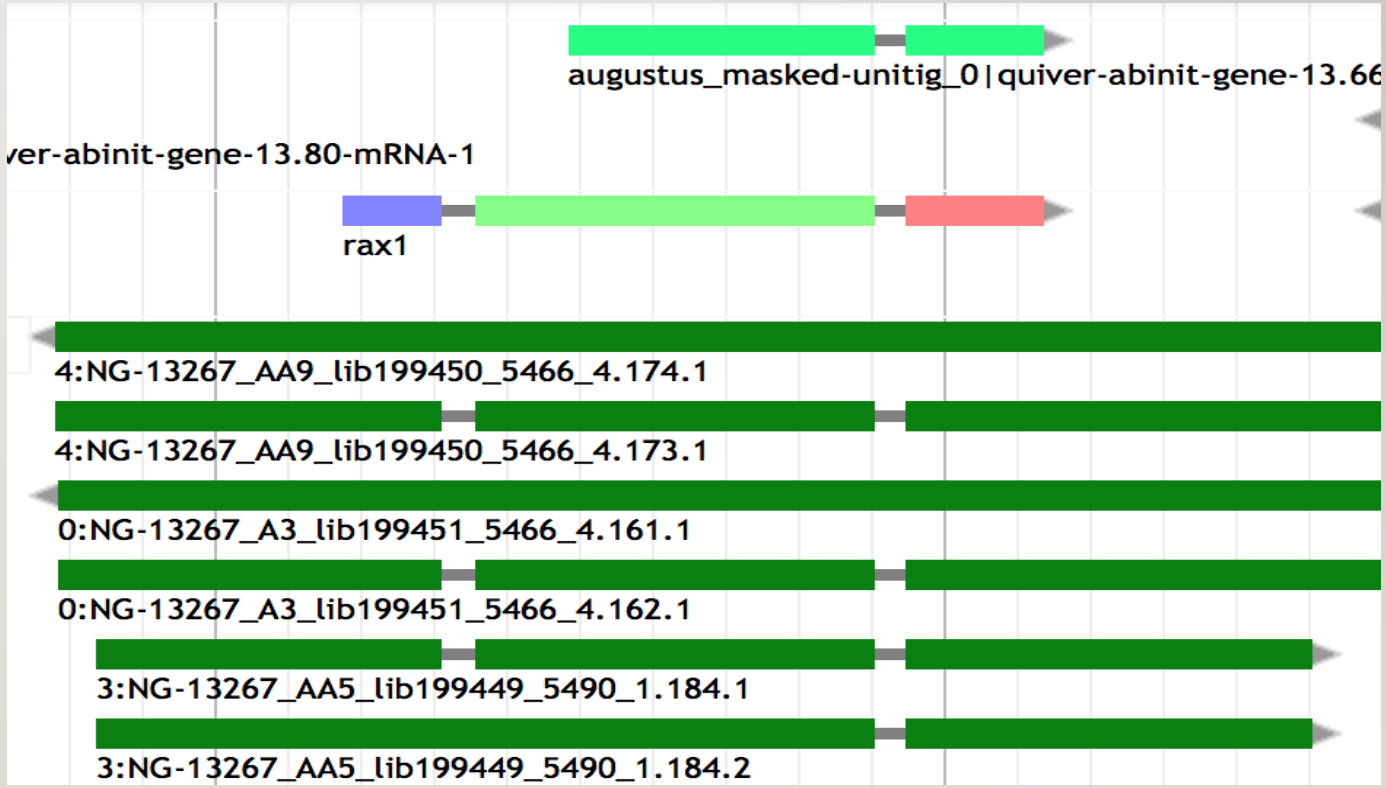
Evidence-drivable gene predictors / Ab initio evidence-based approaches incorporate hints in the form of EST or protein alignments to increase the accuracy of the gene prediction.

Strength :

- Lot of data available
- Protein well conserved
- High accuracy

Limits :

- **Extra computation to generate alignments**
- **heterogeneous sequence quality** :
  - Incomplete
  - Error during transcriptome assembly
  - Contamination
  - Sequence missing
  - Orientation error



Popular tools

	• **GenomeScan**	Blast hit used as extra guide
	• **Augustus**	16 types of hints accepted (gff): start, stop, tss, tts, ass, dss, exonpart, exon, intronpart, intron, CDSpart, CDS, UTRpart, UTR, irpart, nonexonpart.
Self training!	• **GeneMark-ETP**	Protein and transcript based evidence hints
	• **SNAP**	Accepts EST and protein-based evidence hints.
	• **Gnomon***	Uses EST and protein alignments to guide gene prediction and add UTRs
	• **FGENESH+**	Best suited for plant

BRAKER2: automatic eukaryotic genome annotation with GeneMark-EP+ and AUGUSTUS supported by a protein database.

Brůna T, Hoff KJ, Lomsadze A, Stanke M, Borodovsky M. NAR Genom Bioinform. 2021 Jan 6;3(1):lqaa108. doi: 10.1093/nargab/lqaa108. PMID: 33575650; PMCID: PMC7787252.

- Does not require pre-trained parameters or a separate expert-prepared training step.
- Good result

Exercise

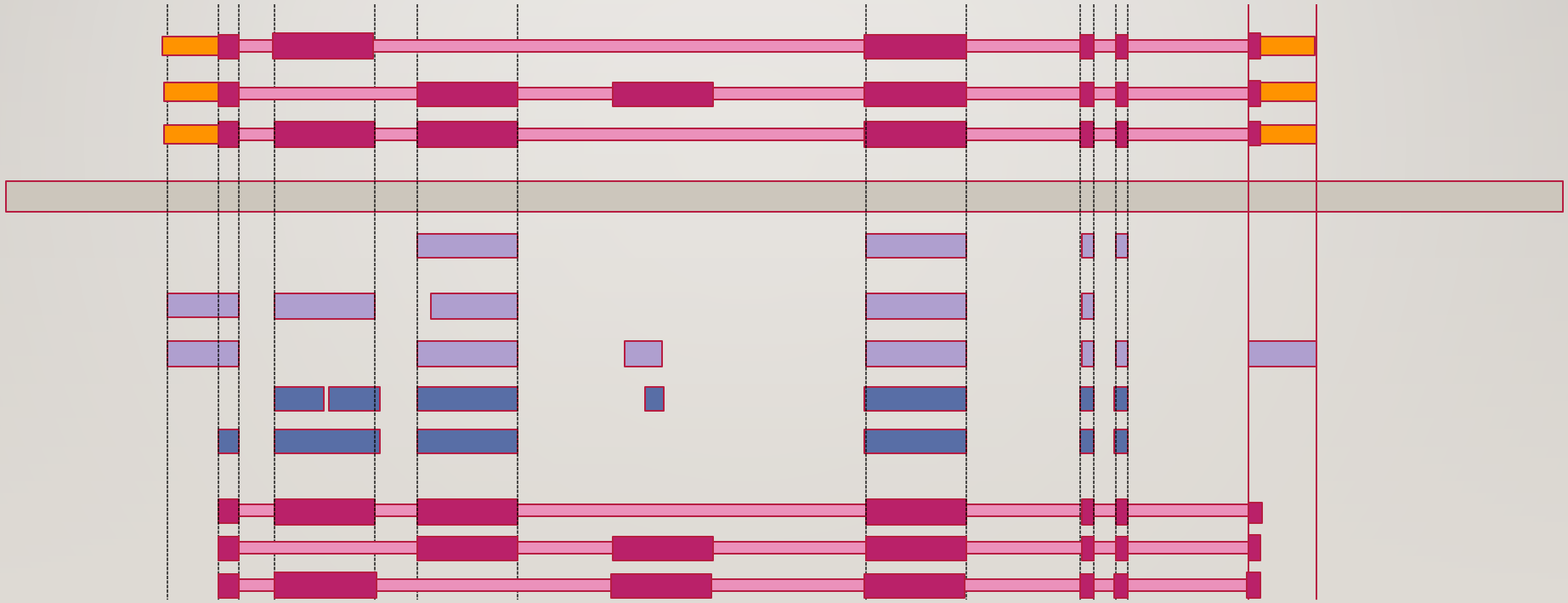
Combiner / Chooser

- *Combining heterogeneous data into gene models*
- *Selection of gene models*

Combiner concept: combining different lines of evidence into gene models

Evidence: ESTs / Transcripts Proteins
Gene prediction (*ab-initio* or evidence-based)
Gene models

=> add untranslated regions (UTR)

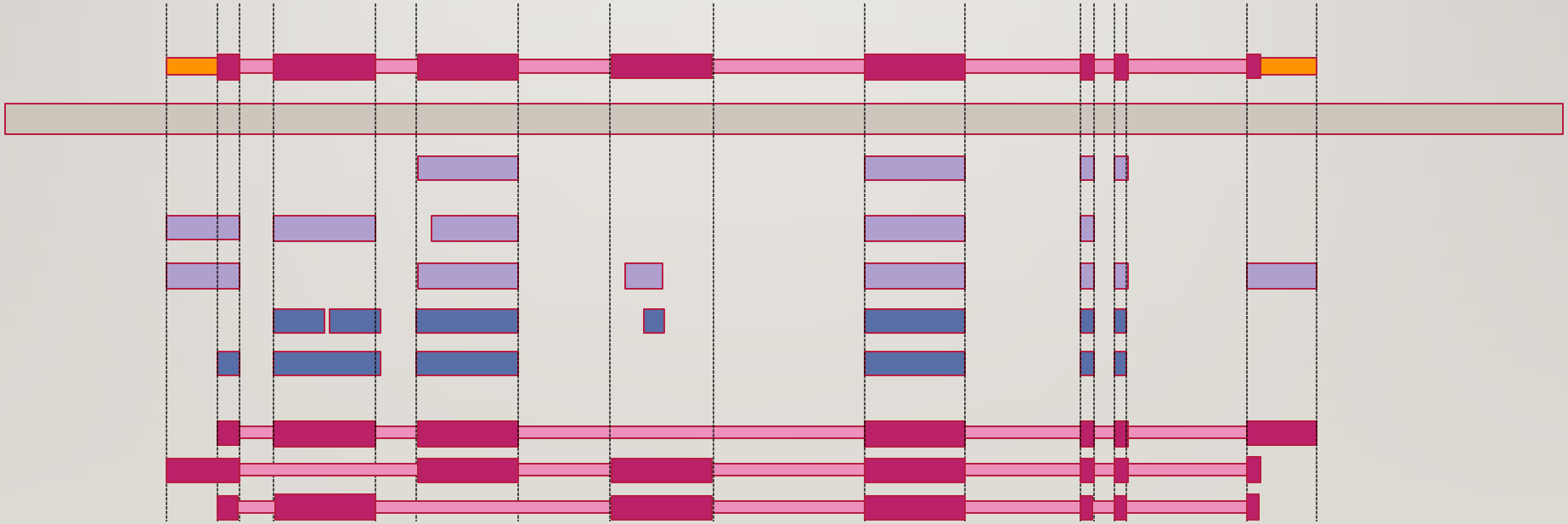


Combiner concept: combining different lines of evidence into gene models

Evidence: ESTs / Transcripts / Proteins

Gene prediction (*ab-initio* or evidence-based)

=> Select the best possible set of exons and combine them in a consensus gene model

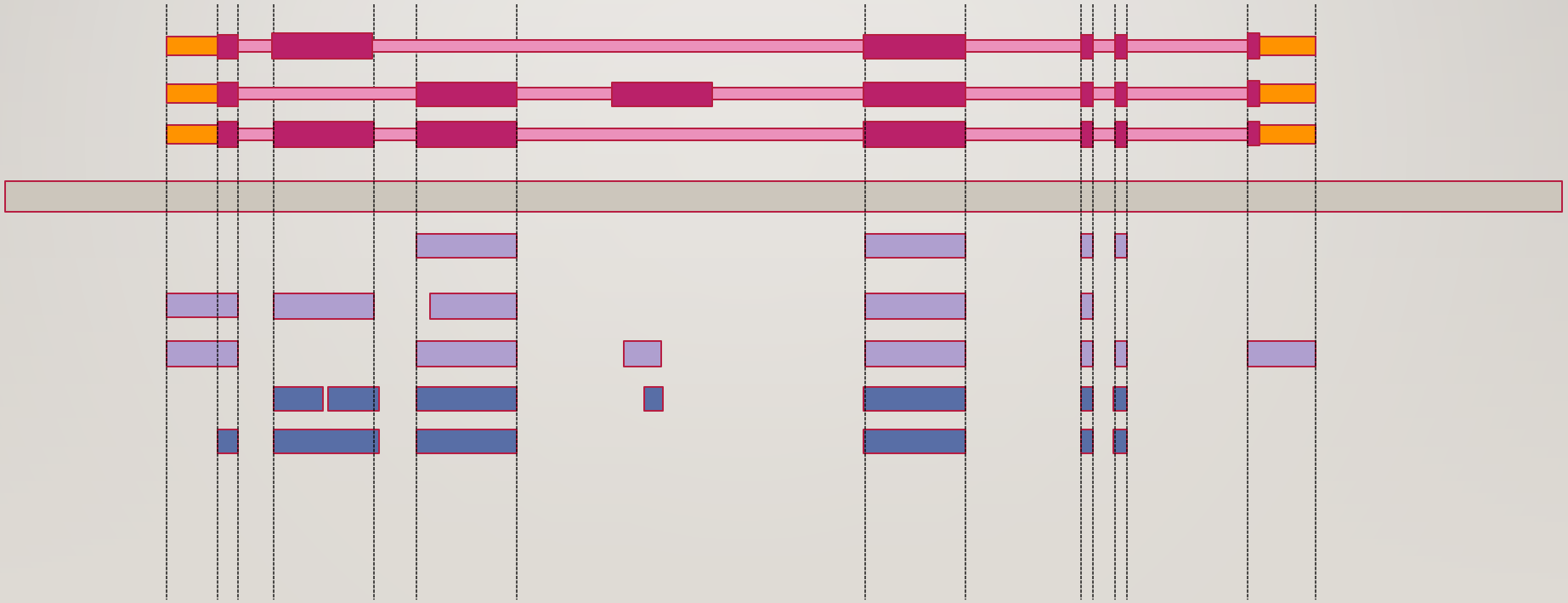


Chooser concept: Select best gene models

Evidence: ESTs / Transcripts / Proteins

Gene models

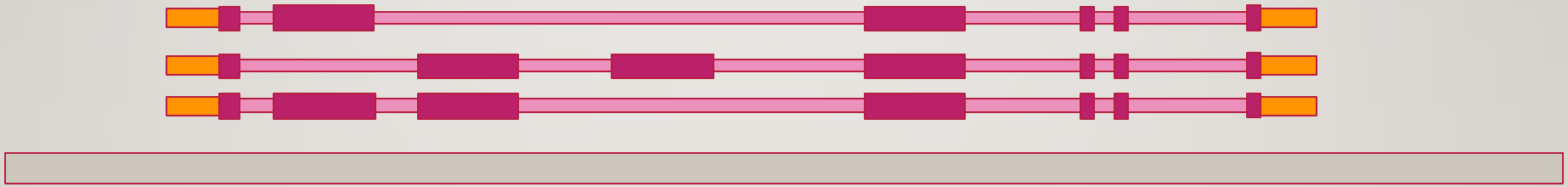
=> Choose the prediction whose best matches the evidence



Chooser concept: Select best gene models

Gene models

=> Choose the prediction whose structure best represents the consensus



Use battery of gene finders and evidence (transcript, protein) alignments and:

Tool	Consensus based chooser	Evidence based chooser	weight of different sources	Comment
A) Choose the prediction whose best matches the evidence				
MAKER*		X		
PASA*		X		
GeMoMa		X		You can weight on presence in multiple genome
B) Choose the best possible set of exons and combine them in a new consensus gene model				
JIGSAW	X			
EVM Evidencemodeler	X	X	X	User can set the expected evidence error rate manually or/and learn from a training set
Evigan	X		X	Unsupervised learning method
Ipred		X		Does not require any a priori knowledge Can also combine only evidences to create a gene model

Strength => They improve on the underlying gene prediction models

Exercise

Pipelines

(The ultimate step)

- *Align evidence*
- *add annotation (UTR, score, gene name)*
- *train ab initio models*
- *Annotation of other features (Repeat, tRNA, etc)*
- *Add functions*
- *And more…*

Tool	Repeat	Transcript / RNAseq	Protein	*Comparative*	*Ab inito*	Function	other	Comment
PASA		X	X					
MAKER3	X	X	X		X		tRNA	Can be used to improve an existing annotation
EGAPx (NCBI)	X	X	X		X	X	tRNA, RFAM	
Comparative Annotation Toolkit (CAT)		X	X	X	X			
Ensembl	X	X	X	X			Rfam Pseudogenes lncRNA	Difficult to use
Funannotate	X	X	X		X		tRNA	Originally for Fungi.
BRAKER3		X	X		X			
EASL		X	X		X	X	BUSCO	Automated with Nextflow

More or less automated…

Exercise

Closing remarks

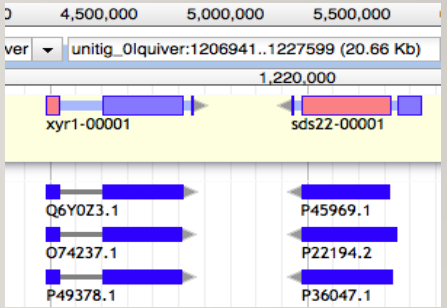
How to choose your Method?

- Scientific question behind (need of a conservative annotation vs exhaustive)
- Species dependent (Fungi / Eukaryotes / Prokaryotes)
- phylogenetic relationship of the investigated genome to other annotated genomes (Terra incognita, close, already annotated).
- Data available (hmm profile, RNAseq, etc…)
- Depending on computing resources (*ab initio* VS pipeline)

Closing remarks

- ❑ >100 annotation tools many methods
- ❑ 6 main class of approaches (Similarity-based, *ab initio*, hybrid, comparative, combiner, pipeline)
- ❑ Many combination of tools

- Several *ab-initio* tools together give better result that one alone (complement each other)
- Pipelines ease the task and give good results
- Most methods only build gene models, no **functional inference**
- No annotation method is perfect, they do mistakes !!
- Annotation requires **manual curation** to be close to perfect
- An annotation (as assembly) is never finished, it can always be improved => **to know how when to stop**
- **QC after annotation** (and before)**!!** BUSCO
- Submit your annotation in public archive **!!**




The End
Thank you for your attention

To go further

Building better genome annotations across the tree of life
https://www.biorxiv.org/content/10.1101/2024.04.12.589245v1.full